

Background

Grounded upon Physics & Mathematics, Cognitive & Perceptual Psychology + Philosophy, unanchored and humbled by the number of stars in the universe - 3×10^{23} (no puns intended, not Hollywood Stars, quite the opposite, I am less than a speck of dust in the cosmic drama)

Vision - Early Years

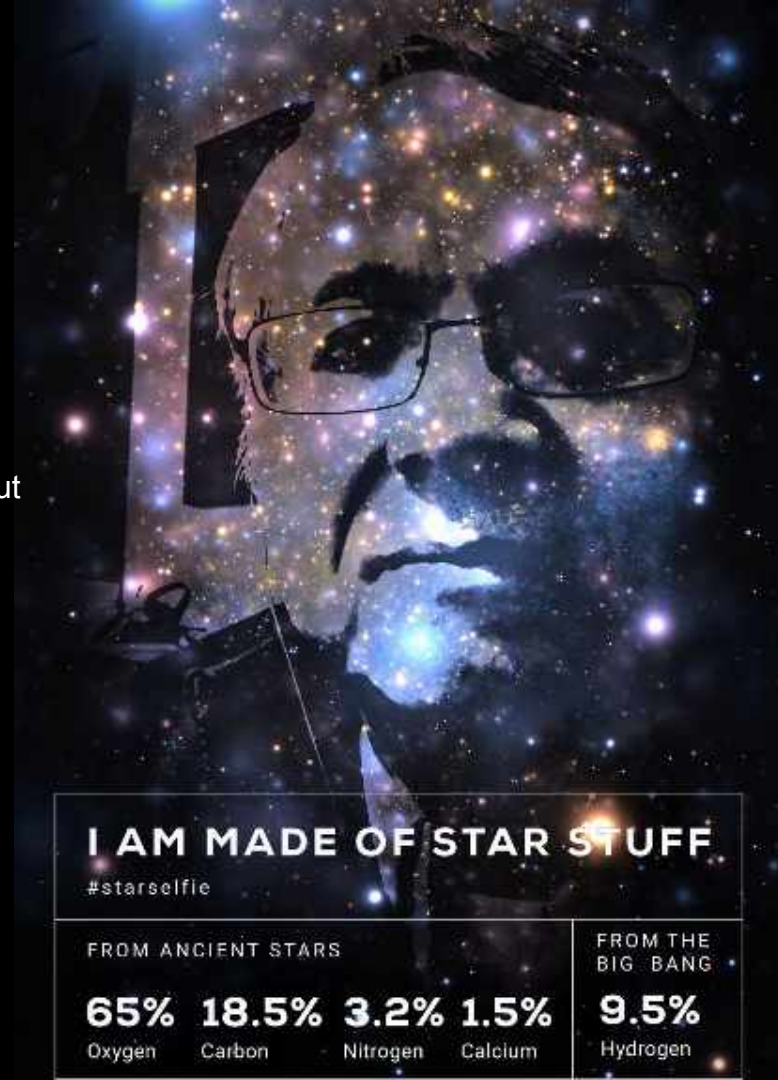
What is reality?

Understand the world within (Mind & Brain) and without (Cosmos/Physical World)

The Next Big Thing

Possibilities...

A device that can sense both our physical and virtual environments on our behalf to accomplish flexible distribution of attention resources (2008, 2019)



I AM MADE OF STAR STUFF

#starselfie

FROM ANCIENT STARS

65% **18.5%** **3.2%** **1.5%**

Oxygen Carbon Nitrogen Calcium

FROM THE
BIG BANG

9.5%

Hydrogen

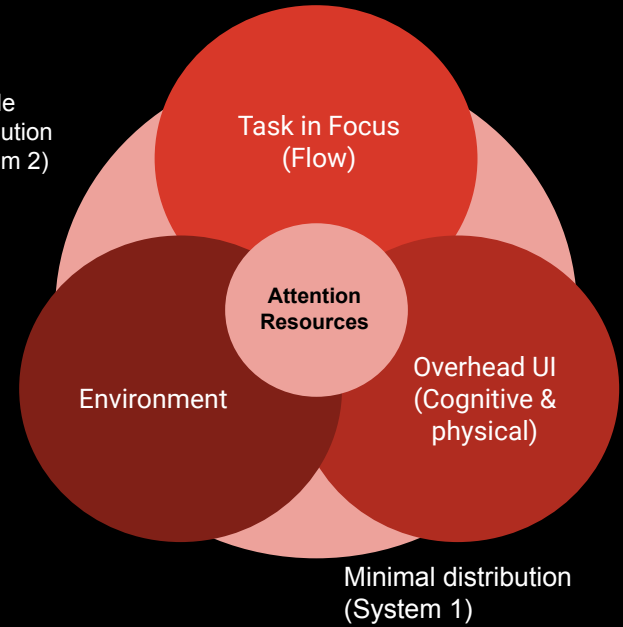
Design Thinking - Flexible Attention

User Interface Design (cognitive and physical) should minimize attention and effort to operate the device (especially in mobile environment).

“Flow is a finely tuned sense of rhythm, involvement, and anticipation. It is a state of “intense emotional involvement” and timelessness that comes from immersive and challenging activities such as software coding or rock climbing.” *Mihaly Csikszentmihalyi*

“A general ‘law of least effort’ applies to cognitive as well as physical exertion.” *Daniel Kahneman*

Flexible
Distribution
(System 2)



Research Thinking - Scientific Discovery & Falsification

Evidence-based, reproducible, repeatable

“Measure what can be measured, and make measurable what cannot be measured” *Galilei Galileo*

“In so far as a scientific statement speaks about reality, it must be falsifiable; and in so far as it is not falsifiable, it does not speak about reality.” *Karl Popper*

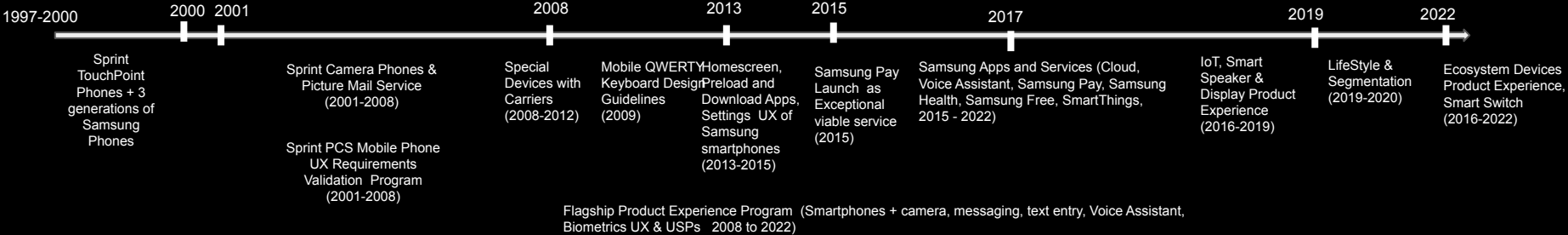
““In science, as in the playing card experiment, novelty emerges only with difficulty, manifested by resistance, against a background provided by expectation.” *Thomas Kuhn*

Being and Becoming: My Journey to Lead UX Design and Research Team

Building UX Team from Total Immersive Product Development Experience:
Founded and expanded Mobile UX Design Innovation and Realization Team (from 4 to 15 members)

Practicing Concurrent & Modular Process for Just In Time and flexible research:
Increases UX team productivity 3X and more & team comradeship (from 2 to 5 + up to 5 research vendors)

Striving for Common Good across different organizations:
Cost Sharing of Qualtrics Platform resulting in greater value for all organizations (across 6 different departments from 2 different divisions)



Building Blocks of High Performance UX Team - A Reflexive Ethnographic Journey

- ❑ Apply UX Design and Research Principles to UX team itself
- ❑ Immersion into workflow to understand goals, work practices, nuances, pain points in relation to the immediate community (cross organization/cross companies). Many of those elements can be ambiguous and emergent.
- ❑ Collect knowledge/experience along the journey and build those into the design of effective, efficient, experience-satisfying and useful structures (roles and responsibilities) and workflow /process (methods and procedures, best practices)
- ❑ Continually seeking feedback and improvement
- ❑ Ideal: optimizing “Flow” work experience for each team member to become expert while having a highly enjoyable work experience.

Additional Principles I learned and abide by as Team Lead of Design or Research

- ❑ Lead by example
- ❑ Maximize learning and career opportunities
- ❑ Prepare for the “Unprepared”
- ❑ Be humble and listen
- ❑ Be dependable and reliable
- ❑ Be adaptable and flexible
- ❑ Diversity & Accessible design

Samsung Laptop / 2-in-1 (2017) sales below expectation

METHODOLOGY

Twelve roundtable discussions were conducted across three geographical locations with premium laptop & design o

This was
evaluat
design.

SESSION

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- Under
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SESSION OUTLINE

FORMAT

- Roundtable discussions
- 3 hour sessions

SESSION OUTLINE

Segment One

- Introduction
 - Discuss Current Behaviors
- Roam and look activity:
We **simulated a typical Best Buy experience** where users could browse the various devices **with all brand logos and trademarks covered**.
- Ranking of devices perceived as top 3:
 - Most Premium
 - Lowest Quality
 - Favorite device
- Detailed device evaluations

Segment Two

- Additional Feature Discussions
 - Brand line price guess
 - Logo style preference
 - Chromebook appeal
 - 2-in-1 Keyboard preference
 - Stylus appeal
 - Fingerprint authorization preference
 - Multiple device keyboard appeal
 - Digital Assistant in laptop appeal
 - Smart home hub appeal

Research identifying root causes of physical design issues impacting on sales of 2017 Samsung Laptop / 2-in-1 (Mobile Computing Product Team, role and responsibility of my team)

Brand agnostic competitive physical design evaluation of laptops in group sessions (with limited quantitative design evaluative ratings to make unmeasurable measurable).

Similar setup has been proven to be effective in our own competitive flagship smartphone research from 2009 through 2016

I negotiated with my manager (and the team requesting this research) for my participation 2ith Mobile Computing Team funding) in exchange for adding MDE research questions (my team was in IoT & MDE team).

Both teams agreed with the protocol and format of the session (segment 1: 2.25 hours for physical design; segment 2: 45 minutes for IoT/MDE questions)

Research in process - a tale of internal team conflicts

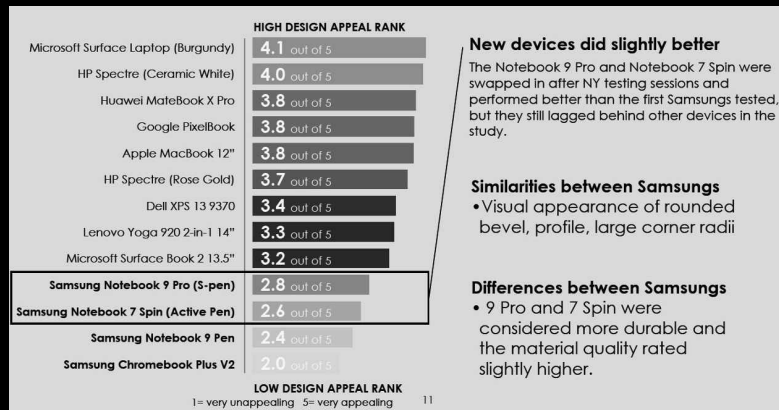


The interesting but sad situation in this project is the conflict between the TX and NJ team members in Mobile Computing Product team (they reported to the same director).

I became their moderator and designed the study merging their requirements and got their approvals and executed the project.

Measurable Findings with Detailed Qualitative Data

All the Samsung devices received bottom overall design scores and lacked characteristics to make them “favorites” or appear “premium”



New devices did slightly better

The Notebook 9 Pro and Notebook 7 Spin were swapped in after NY testing sessions and performed better than the first Samsungs tested, but they still lagged behind other devices in the study.

Similarities between Samsungs

- Visual appearance of rounded bevel, profile, large corner radii

Differences between Samsungs

- 9 Pro and 7 Spin were considered more durable and the material quality rated slightly higher.

Consumers described Samsung devices as “Boring”, “Basic” and “Toy-like” while their Favorite and High Quality Devices as “Sleek”, “Modern” and “Clean”

“The appearance is very basic and clunky.”

Notebook 9 Pen

“It looks old - like 1987-old.”

Notebook 7 Spin

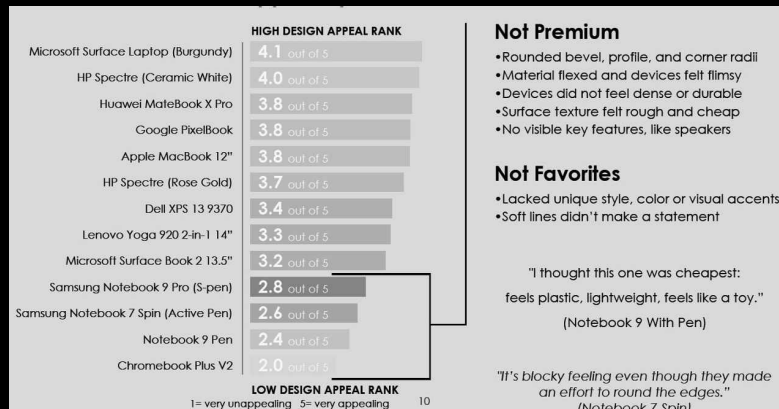
“Nothing stands out.”

Notebook 9 Pro 13”

“Looks like an old flip phone.”

Notebook 9 Pen

Clunky
Toy-like Boring
Basic Outdated
Functional



Not Premium

- Rounded bevel, profile, and corner radii
- Material flexed and devices felt flimsy
- Devices did not feel dense or durable
- Surface texture felt rough and cheap
- No visible key features, like speakers

Not Favorites

- Lacked unique style, color or visual accents
- Soft lines didn't make a statement

“I thought this one was cheapest: feels plastic, lightweight, feels like a toy.”
(Notebook 9 With Pen)

“It's blocky feeling even though they made an effort to round the edges.”
(Notebook 7 Spin)

“Very sleek and elegant look.”

Microsoft Surface Laptop

“It's clean and sleek looking. You can spot the laptop anywhere.”

HP Spectre - Rose Gold

“Sleek, modern, nicely designed.”

Google Pixelbook

“Looks expensive. Modern.”

HP Spectre - White

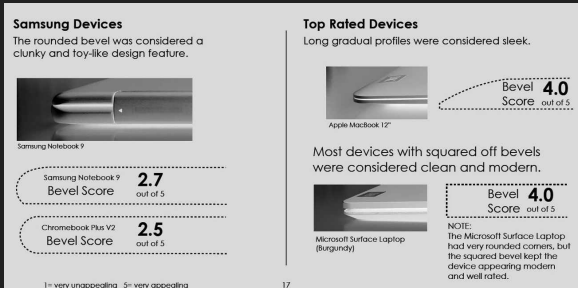
“It feels just right. It feels smooth and sleek and compatible.”

HP Spectre - Rose Gold

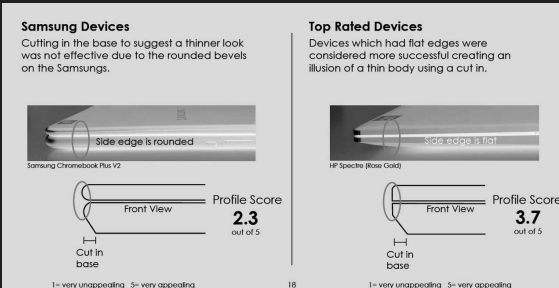
Stylish
elegant Modern
Sleek Functional
Clean

Evidence-driven design recommendations

Bevel

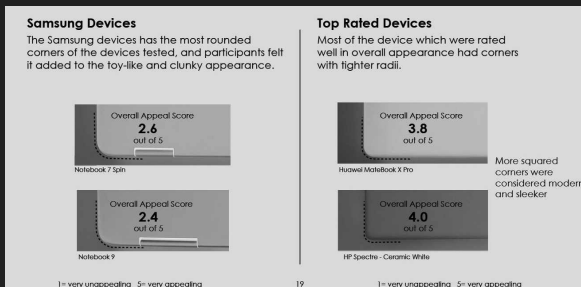


Profile Taper

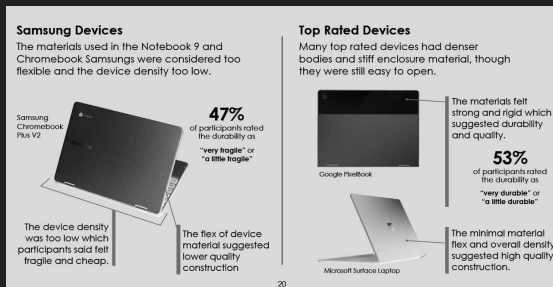


The findings and recommendations were accepted by HQ Design Center.

Corner Radii



Perception of Durability



The redesign took place in forthcoming generations and enabled that team to accomplish more than 100% of their sales goal after the new design was launched into the market in 2019.

Key Takeaways and Learning

- ❑ Uphold UX research value beyond organizational dynamics and personal ego, but be sensitive to team and organizational conflicts
- ❑ Being neutral and impartial with evidence driven research is one to the best ways to navigate in a complex and challenging environment
- ❑ Seek for the common good both within and between teams and be tolerant of differences as in an open society

Thank you all for giving me time sharing my experience!

I have also prepared an extended appendix with 2 project examples illustrating the importance of UX research and my detailed research experience. If you would like to review those examples or my detailed research experience, please send me a message and I will share those information with you..

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Appendix

Research Experience

Galaxy Home (2018-2019)

Samsung Pay (2015)

Research Thinking (2019)

Product experience research - Methodology. Tools, Skills

Type	Product / Service	Methodology
Mobile Products (Devices) & Ecosystem	Special devices with Verizon & T-Mobile, Flagship Galaxy Smartphones (GS1 through GS22, Note series & Z Fold and Z Flip), Wearables (Galaxy SmartWatches, Earbuds), IoT Devices (Smart Speaker/Display, OBDII Hotspot, Smart Home Products), Tablets, Laptops/2-in-1 (2008 - 2022) Scope: Physical design and ergonomics, OOB, core apps UX, cross-device UX, packaging mostly hands-on evaluation	Design: Qualitative Participatory Design group sessions with consumers using paper and art supplies or walking through different competitive products. Physical Design: Quantitative: Multilinear, PLS Regression; Central Composite Design Summative & Competitive Benchmark (Mixed Methods: Quantitative - t-test, ANOVA, Chi-Square, Semantic Differential, Kruskal-Wallis test, Friedman test, multiple comparisons; Qualitative - User Diary, Ethnographic Interviews (in simulated environment), Online Panel, 1:1 interviews, Video Diary, Focus Groups; Large scale user trials Formative & User Trials (Mixed Methods as above) Quantitative Survey (mostly for measurable scores like SUS or UEQ)
Mobile Services / Apps	Samsung Services & Apps: Media Hub, Samsung Account, Samsung Pay, Samsung Health, Smart Switch, SmartThings, Samsung Free (2010 - 2022) Scope: service/app UX design and performance	Formative and Summative Service or App Experience, Task-based usability, competitive benchmark & app performance (Mixed Methods: Similar to Mobile Products) Quantitative Survey (mostly for measurable scores like SUS)
Settings & Homescreen Organization, Core Apps, Preload Apps and Downloaded Apps	Communication apps (Call, Messaging, Video Call), Voice Assistants, Camera, Settings, Homescreen, Email, Internet, Notes, Text Entry Methods, (2008 - 2022) Scope: In depth UX Review, Interaction Design, Visual Design and Information Architecture and performance	Formative and Summative Product Experience, Task-based usability, competitive benchmark & app performance (Mixed Methods: similar to Mobile Products with the addition of Card Sorting, Icon Discrimination & Understandability, Task-based evaluation of Information Architecture, e.g., Optimal Workshop Quantitative Survey (mostly for measurable scores like SUS or UEQ)
Marketing	Product features prioritization/trade-off; pricing models (flagship smartphones, bundled device/service e.g., music services, mid-tier /value tier smartphones, laptops/2-in-1, accessories (2009-2020)	Conjoint Analysis, Kano Model, van Westendorf Pricing Model, Product Concept Scale (proprietary)
Consumer LifeStyle, Attitudes and Behavior	Consumer Smartphone Segmentation (2019) Elements of Value: Loyalty and Product/Service Satisfaction (2020) Ecosystem and Multi-Devices Usage & Behavior (2022)	Mostly using Quantitative Surveys (Clustering, Multivariate Regression - Linear, Logistic, PLSR, Chi-Square, t-Test, ANOVA, Multiple Pairwise Comparison) Limited qualitative feedback is also included in most surveys. Textual analysis is performed either manually or through AI driven analysis (e.g., Text IQ from Qualtrics)

Failure to Launch - Galaxy Home Speaker Fanfare

Announced in Samsung UnPacked 2018 (Aug 9 2018)

<https://www.youtube.com/watch?v=1lw8fe0dtlq>

Galaxy Home USPs

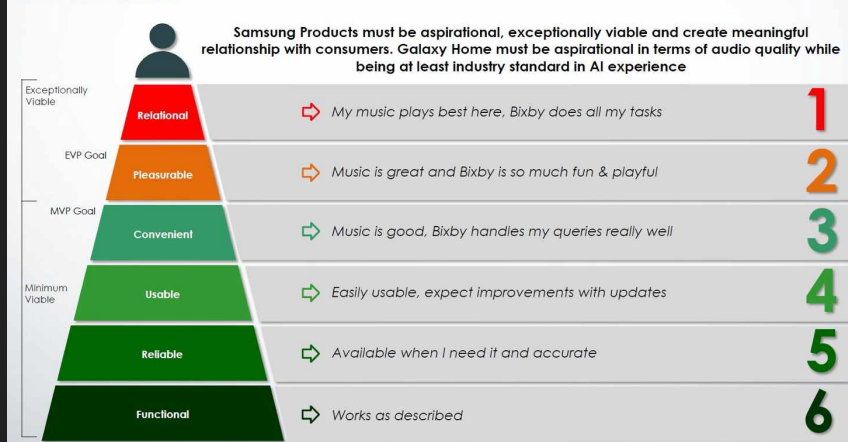
Sound Quality	Pass
AI Assistant - Bixby	Fail
Smart Home Hub	Pass
Better Together with Galaxy Smartphones	Pass
Better Together with other Samsung Products	Pass

Samsung co-CEO DJ Koh said in Feb 2019 that the Galaxy Home would go on sale by April.

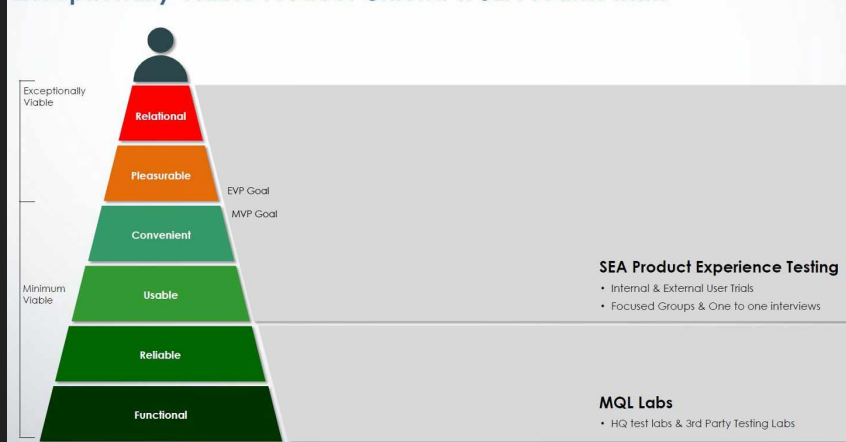
“Two years later, and Samsung Galaxy Home still hasn’t shipped. What happened?” –Alistair Charlton, gb gear brain, Aug 13 2020

The research done by my team showed that Galaxy Home did not meet minimum viable product criteria. I also nudged HQ Quality Assurance using our rigorous methodology with full utterance set (2011) and multiple speakers (3) as their functional test.

Product Goals



Exceptionally Viable Product Criteria & SEA Teams R&Rs



Methodology



Galaxy Home
Google Home Max
Amazon Echo Plus
Apple HomePod

124 utterances
from 12 categories
(required by HQ
Usability Team)

1 recorded voice (with
inflection and volume
control) used to test
across all 4 smart
speakers

Richardson TX
MDE Demo Room
Controlled Setting with
little noise disturbance

Male native speaker of
American English, age
20-29



Criteria Definitions

1. How many times before the AI woke up?	Did the AI wake up when its wake up phrase was called? How many times did it take?
2. Was the response time appropriate for a regular conversation?	Did it respond in a timely manner? Yes – Didn't feel delay compared to competing devices No – Delayed response
3. Did it answer?	Did it answer the utterance?
4. Did it understand?	Did understand /not understand the utterance ?
5. Did it answer as expected?	Did the AI give the expected response to the utterance?
6. Was the response natural?	Does it feel natural response, without robotic stuttering/awkwardness almost like a human gave the response?

Utterances were played from a JBL 300 speaker (speaker volume = 98%) to all smart speakers with the same male recorded voice from the same distance (3.5 feet).

A sound meter was also used to verify sound intensity at each smart speaker was the same.

Recorded wake word was played to the corresponding smart speaker followed by the recorded utterance. The same 124 utterances were played to each speaker (using AutoHotkey to speed up the testing)

Results were rated with 6 criteria.

Results

Overview	Wake up success %	AI Acknowledgement (within 1 second)	Did it answer the utterance? (any answer)	Did it understand the utterance?	Did it give the expected answer?	Was the response natural?
Bixby	93%	77%	87%	29%	23%	60%
Google	99%	100%	100%	94%	92%	98%
Alexa	100%	98%	98%	81%	78%	94%
Siri	100%	98%	96%	73%	73%	90%

Accidental wakeup example



Bixby woke up **24** times unaddressed during the 4 hour experiment, with mostly irrelevant answers.

All other AI speakers did not wake up unaddressed times.

Irrelevant response example

Hi Bixby, tell me about the band Metallica.

Sure, there are several different locations you can get your drivers license...



Fail to Launch Galaxy Home - Sad but needed

It is a testimony to the importance of scientific experimental design in UX research. Not even the top executives of our company could dismiss the experimental design and the results from this research, despite their repeated public announcement of imminent launch of the product. In the smart speaker market in 2018, we will need an exceptional viable product to compete with Google, Amazon and Apple and unfortunately, Galaxy Home was below our criteria of minimum viable product.

Complex organizational and cross-cultural dynamics made the terrain difficult and treacherous to navigate. However, as researchers, if we stay true to our course and demonstrate the scientific basis and value of our research (and design) rigor, with harmonious relationship with other organizations, we can effectively establish and reinforce the role of UX research across all phases of product development.

We ran a pilot using a more rigorous methodology in controlled environment with results shared with their team. The MQL team accepted the changes and completed their testing on functional and reliability aspects of Galaxy Home. The results corroborated with our pilot and the rest was history.

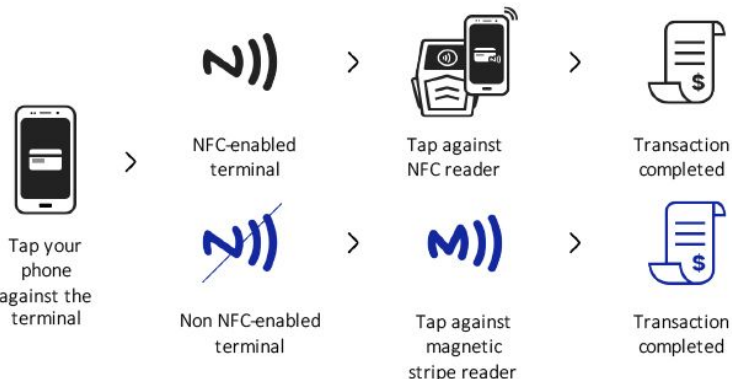
Results were further validated by Quality Assurance Laboratory with our methodology and criteria. We provided the template to record the results for them. They extended the process using 2011 utterances with a female native American English speaker and a male non-native American English speaker as recommended by my team. Despite more attempts had been made to improve Bixby performance, the results did not meet Minimal Viable Product criterion as of June 2019.

Launching Samsung Pay as exceptional viable service

	 Apple Pay	 Samsung Pay	 Android Pay
Supported Devices	iPhone 6, iPhone 6 Plus, Apple Watch	Galaxy S6, S6 Edge, S6 Edge Plus, Galaxy Note 5	All NFC-enabled Android devices
How to Use	Tap to pay with NFC at supported terminals	Tap to pay with NFC at supported terminals, also supports mag-swipe readers	Tap to pay with NFC
# of Accepted Locations	700,000	> 30 million	> 700,000
Notable Retailers	Walgreens, Whole Foods, Toys 'R Us, Staples, Panera, Subway, McDonald's, Macys, Aeropostle, Disney Store	Majority of credit-card accepting retailers	McDonald's, GameStop, American Eagle, Best Buy

Samsung Pay (2015) had to compete with a well-trenched market leader Apple Pay which launched a year earlier. In order to survive competition, Samsung Pay offered a unique competitive advantage - the use of magnetic secure transmission (similar to using magnetic strip technology in credit card). Apple Pay made use of NFC only and was accepted only in few retailers. Samsung Pay supported both MST and NFC payment.

It would take another 3 years for most POS systems to support NFC payment



The research done by my team was an example of a massive project undertaken by our larger team with our roles and responsibilities. The cross-functional composition of the launch team included Samsung Pay Business team in CA and R&D teams in CA and MA, two product managers from our Product Planning team (Content and Services), 5 UX researchers (including myself running formative usability evaluations and my senior manager running user trial). The entire research project took about 9 months to complete. The project ran smoothly with almost no cross-functional challenges with our detailed plans and preparation. The only challenge we faced was scheduling of facilities for the research and some delays of the availability of working prototypes for evaluations.

Launched in August 2015, Samsung Pay remained as the most successful service Samsung ever launched into the market. As of today, Samsung Pay received 4.5 stars with over 1 million reviews and 100+ million downloads in Google Play Store.

*Note that Samsung dropped MST support in 2020 as most POS terminals (and retailers) supported NFC.

Launching Samsung Pay as exceptional viable service

Phases	Methodology, Sample, Devices used	Tasks	Impact
Alpha Field Trial (Mar 2015)	Field testing (retailers including convenience stores, big box stores and stores in malls) done by Samsung Pay Product team and SEA Product Experience Research	Used Samsung Pay in as many retail stores in a week for payment and refund process and reported in store experience with both the payment process and reaction from store employees.	Many retailers were surprised to learn Samsung Pay could be used in their POS (Apple Pay did not support MST). A few did not allow transaction to be made. Results informed Marketing of working with retail business to publicize Samsung Pay capability
Formative Usability Evaluation (5 waves: Mar 2015 - Sep 2015)	Mixed methods: both payment system performance and user quantitative/qualitative feedback were captured. Consumers and Internal employees. Number of participants (19 to 25 each wave) Prototypes, both POS systems (4 terminals tested), supporting smartphones and Pay apps (Samsung & Apple)	Iterative task-based evaluation covering UX - OOB, setup, app invocation and use for payment transactions, User Journey: Setup (Samsung Account + Fingerprint), Add card(s), NFC and MST Transaction, Refunds, Card Status change, Card lost/Stolen and Deregister Competitive Pay app (Apple Pay) was used as benchmark.	Provided immediate feedback to R&D and Design team on critical UX issues related to POS system, devices and Samsung Pay app covering both hardware and software. The critical UX fixes and improvements were pivotal to the highly successful launch of Samsung Pay in the market (August 2015 with GS6 Edge+ and March 2017 with all models of GS7) as an exceptional viable product. (4.5 star over 1 million reviews and 100+ million downloads as of today)
Beta Field Trials (Jun - Sep 2015)	Consumers, employees (N=1200) in 12 cities	Front-end UX evaluation after each transaction (d-Scout as tool)	Back-end transaction data were analyzed by R&D to identify error and issues. Validation of Samsung Pay UX and performance (both front end and back end) at massive scale.

[illegible]

- Participants were observed unlocking the device, then failing to authenticate with fingerprint immediately after on repeated occasions, indicating additional room for improvement
- Issues were not UI issues, but hardware / software

Recommendation

- Continue to improve upon fingerprint authentication within Samsung Pay and overall to compete with LO.

Recommendation

- Work with Square to modify message on screen to advise user making payment with phone that they should hold their phone in place for up to 5s.

The diagram illustrates the app launch sequence with three screenshots of an iPhone. The first screenshot shows the lock screen with the time 11:18 and a date of 11/11/11. An arrow labeled '2.5s' points to the second screenshot, which shows the app icon on the home screen. Another arrow labeled '1.0s' points to the third screenshot, which shows the app's splash screen. Above the third screenshot, a table summarizes the timing:

	Time (s)	Time (s)
Time to launch app	6.0s	2.5s

Recommendation

- If possible, improve speed with which the app launches to 1-2 seconds.

Terminal	Distance	Device Orientation	Time to Register
Customer Facing (Webcam Model)	<15"*	On the side	<3s
Counterfeit vs PIN (Ingenico Model)	<0.5"	On the side, next to magnetic strip	Up to 4s
Portability: M65 (Ingenico Model)	<0.5"	On the side, next to magnetic strip	Up to 4s
Square	<0.5"	Directly on Square	Up to 5 s

Recommendation

- Focus marketing and educational material on the following:
 - Place phone on the side of terminal where you slide card
 - Keep in place for a few seconds
 - Place directly next to reader
- Customers should not be asked to "tap", seen in some tutorials.

Improvements in fingerprint accuracy and MST signal strength contributed to the increased success

Fingerprint Success	Build 06.01 (m=1)	Build 06.01 (m=2)
Samsung MST (Tx)	30%	42%
Samsung MST (Tx)	52%	65%

*Success estimated when testing 06.00.00, as discussed on 05.25.20. Success had to be 60% or above for the comparison to high failure rates with MST only.

Recommendation

- Continue to improve fingerprint authentication within the app, MST signal strength, and provide marketing and education materials focusing on how / where to position the device when making payments.
- Swiping a credit card is successful ~90% of the time; in order to achieve viable replacement, Samsung Pay should strive for that level of success.

MST Distance Build 06.01

MST Distance Build 06.04

Terminal	Distance	Device Orientation	Time to Register
Customer Facing (Hygiene Module)	<1.5"	On top of NFC reader	<1s
Countertop in FNN (Hygiene Module)	<1.5"	On top of NFC reader	<1s
Portable (Hygiene Module)	<1.5"	On top of NFC reader	<1s

LO Pay Active Area (4" - 6")
= 2x more range than Sampling Pay

NFC Reader Center

Sampling Pay Active Area

Recommendation List	Rank 1 (%)	Rank 2 (%)	Rank 3 (%)	Rank 4 (%)
A1	2%	17%	66%	15%
B1	22%	33%	48%	10%
C1	10%	20%	38%	32%
D1	17%	46%	24%	13%

Source: "Rank 1" Items "C1" from user

Legend:
 Rank 4 (light blue)
 Rank 3 (medium blue)
 Rank 2 (dark blue)
 Rank 1 (darkest blue)

Images shown below the chart:
 A1: A yellow smartphone with a blue screen.
 B1: A white smartphone with a blue screen.
 C1: A black smartphone with a white screen.
 D1: A black smartphone with a white screen.

Failure caused by fingerprint authentication reduced significantly, but still contributes to 25% of error Activation of Square camera caused increase in 'Device Not in Position' error

Reasons for failure (Device Not in Position)	Build 4.04 (n=15)	Build 4.04 (n=24)
Fingerprint	42%	20%
Device in Wrong Place	20%	20%
Device Too Far Away	24%	29%
Angle Wrong	2%	1%
Device Not in Position Long Enough	7%	11%
Unrelated error	NA	3%

Device Too Far Away
29.4%

Device in Wrong Place
28.6%

Fingerprint Doesn't Authenticate – 24.6%

Additional details on each reason for failure analysed on following slides

15s

12s

Recommendation

- Explore idea that payment processor can detect payments made by tokenized card and communicate back to terminal that additional security isn't required (PIN or signature).
- Ensure last four digits of the virtual card display on the card.
- Finally, determine how often this information is requested through field trials.

Research Thinking - Mindset (2019)

Research: Evidence-based, reproducible, repeatable

- ❑ Willing to experiment, take risk and fail
- ❑ Pay your due with real hands-on experience; Learn how to do E-2-E user research – Plan and run your own research, analyze and interpret data yourself
- ❑ Learn from professional researchers from different agencies; improve upon their methods
- ❑ Driven by curiosity and urge to learn and understand
- ❑ Research rigor is harder to find than “research” insights

“Our knowledge can only be finite, while our ignorance must necessarily be infinite.” *Karl Popper*

*extracted from the Design/Research Workshop for HQ CX Insights teams from Korea, Europe, China and US (April 2019).

Research Thinking - Lesson Learnt (2019)

- ❖ Paper prototyping exercises can work magic in early design
- ❖ Active participation (not the type in traditional focus group studies) is essential for all phases of product design and usability research
- ❖ Combine different methodologies in planning and execution of single research sessions to keep participants engaged
- ❖ Use the largest number of participants you can afford and analyze your pilot carefully
- ❖ Use realistic simulation (it is like having participants to be actors in a theatrical play) to capture needs and usage
- ❖ Make use of every opportunity to learn and run research related to topics of interest in parallel to primary research undertaken by vendor (e.g., using floaters in primary research or recruiting more participants to run parallel sessions)

*extracted from the Design/Research Workshop for HQ CX Insights teams from Korea, Europe, China and US (April 2019).